Curriculum

Data Science And Analytics



Module	Topics Covered
Introduction to Data Science	What is Data Science, Data lifecycle, Tools overview (Python, Jupyter, Colab), Career roadmap
Python for Data Analysis	NumPy, Pandas – DataFrames, Series, data manipulation, file handling (CSV/Excel)
Data Collection & Preprocessing	Web scraping basics, handling missing values, outlier detection, data cleaning techniques
Exploratory Data Analysis (EDA)	Descriptive statistics, correlation, distribution analysis, data storytelling
Data Visualization	Matplotlib, Seaborn, Plotly – charts, heatmaps, dashboards
SQL for Analytics	SQL queries, joins, aggregations, subqueries, real-world business queries
BI Tools (Power BI/Tableau)	Connecting data, building dashboards, KPIs, publishing reports
Capstone Mini-Project	End-to-end dataset analysis with EDA + dashboard presentation